

smartphone has now. This means, new engine-management functions and security features would be just a download away. These software updates in cars even add features like automatic emergency braking and blind-spot warning.

Engine management. What is more interesting is that, in some cars like Tesla, these software updates can increase acceleration, economy or top speed of your car after it gets installed. Expect a day in the future when you can buy car-performance upgrades through app-like downloads, instead of going to the workshop to get the engine modified.

Convenience. Our own *desi* company, Mahindra Reva Electric Vehicles (MREV), has an amazingly connected electric car in the guise of e2o. Users are able to control various elements in their car through a smartphone from anywhere in the world—from turning on the AC to pre-cool the car before you take your family out on a hot summer afternoon, to remotely locking the car and even charging it.

An interesting feature is REVIVE, which enables the driver to use the smartphone to activate an additional reserve of eight kilometres of range if the car runs out of power.

Safety. Chevrolet is believed to be planning to use its onboard 4G LTE Wi-Fi and OnStar RemoteLink smartphone app to alert owners if their car is about to break down.

Another example is when National Highway Traffic Safety Administration made two recall announcements. While this would normally require owners to bring in their cars to get it fixed, Tesla was able to solve the problem via a simple software update.

Cars can now talk to each other

Perhaps, the first bit towards a world with friendlier cars would be when these can start talking to each other.

Major firms are already running with the idea. GMC already has plans to begin installing vehicle-to-vehicle

(V2V) communication in every GMC car.

Nokia announced a US\$ 100 million Connected Car fund in May 2015 to drive innovation in connected and intelligent cars.

The month of May also saw the US transportation secretary make an announcement; a mandate proposing all new cars and trucks to have V2V communication equipment will be proposed to the government earlier than expected.

Back in 2006, V2V and vehicle-to-infrastructure (V2I) communication using dedicated short-range communications (DSRC) based on wireless local area network (WLAN) technology were tested by what was known as Vehicle Infrastructure Integration (VII) initiative.

The Federal Communications Commission has dedicated the 5.9GHz frequency band for automotive communication between vehicles and roadside units (RSUs).

IEEE 802.11p is a standard to add wireless access in vehicular environments (WAVE), thus enabling the support of intelligent transportation systems (ITS) applications. It is an amendment to IEEE 802.11, which is the same as the Wi-Fi we all use in our homes and phones today.

“V2V communications is still not mainstream with the Internet of Things (IoT), since many car manufacturers are contemplating the risks associated with it. However, some car manufacturers such as Volkswagen and Audi are working on V2V features for their own cars, such that one Volkswagen may not talk to all cars around it but all Volkswagen around it,” explains T. Anand, managing director of Knewron.

He adds, “For all cars to talk to each other, either single protocol or language has to be agreed upon (which is another debate topic for all things in the IoT) or, alternatively, all communication must pass through a broker or translator, which will then make cars another thing in the IoT.”

Inter-Vehicular Network Technol-

ogies (INVENT), led by Grace Wang from New Jersey Institute of Technology (NJIT), the USA, is developing computing, sensor and networking technologies for next-generation vehicles.

These could gossip, too. These connected cars could very well do the automotive flavour of gossip. This means that these would be constantly swapping basic and minor data that would become useful when a related event occurs, while at the same time giving us an idea of how vehicles in the vicinity are behaving.

Sensor networks and Big Data. Sensors are a very big part of this game. An example is a sensor network across roads, somewhat similar to the ones surrounding some secure areas of Area 51 (a remote detachment of Edwards Air Force Base, within the Nevada Test and Training Range, the USA), which detects motion along the area.

In our case, it could be used to detect animals crossing the road and alert incoming cars to slow down, accordingly. Now, if a vehicle brakes hard, it can use its V2V network to alert other cars in the same lane to slow down urgently; technology like this could go a long way in preventing pileups.

Traffic re-routing and avoiding congestion on the road are other benefits with technology like this. Currently, smartphone users are able to detect congestion along a stretch of road by using applications like Google Maps. Dynamic traffic assignment (DTA) algorithms are examples of technology that could enable people to reduce travel time. However, these lack scalability and robustness and require higher computation time.

The INVENT team also proposed road based vehicular traffic (RBVT) routing, which uses real-time traffic data from vehicles and roadside sensors like the ones mentioned earlier.

Interference in the spectrum. Scott Belcher, chief operating officer of the advocacy group Intelligent Transportation Society of America, stated in a news report on Voice of America